Algebraic Fractions

Algebra · Edexcel IGCSE Mathematics A (4MA1) Paper 1 · Year 10 · worksheet

A+ Maths Hub

Name	Class	Date

Answer ALL questions.

Write your answers in the spaces provided. You must write down all the stages in your working.

Suggested time: 103 minutes · Total marks: 103

KEY RULES

Simplify	Factorise top and bottom, then cancel common factors
Multiply	Multiply tops and bottoms; cancel before multiplying
Divide	Flip the second fraction and multiply
Add / subtract	Use a common denominator
Equations	Multiply every term by the common denominator

WORKED EXAMPLE 1

[3 marks]

Simplify $\dfrac{x^2-9}{x^2+x-6}$

Step 1	Factorise the top (difference of two squares)	$(x-3)(x+3)$
Step 2	Factorise the bottom	$(x+3)(x-2)$
Step 3	Cancel the common factor $(x+3)$	$\dfrac{x-3}{x-2}$

Answer: $\dfrac{x-3}{x-2}$

WORKED EXAMPLE 2

[3 marks]

Write $\dfrac{2}{x+1}+\dfrac{3}{x-2}$ **as a single fraction.**

Step 1	Common denominator	$(x+1)(x-2)$
Step 2	Write each over it	$\dfrac{2(x-2)+3(x+1)}{(x+1)(x-2)}$
Step 3	Expand the top and simplify	$\dfrac{5x-1}{(x+1)(x-2)}$

Answer: $\dfrac{5x-1}{(x+1)(x-2)}$

A+ Maths Hub International Community Schools

WORKED EXAMPLE 3

[3 marks]

Simplify $\frac{x}{3} \div \frac{2x}{9}$

Step 1	Flip the second fraction and multiply	$\frac{x}{3} \times \frac{9}{2x}$
Step 2	Multiply and cancel x	$\frac{9}{6}$
Step 3	Simplify	$\frac{3}{2}$

Answer: $\frac{3}{2}$

1 Simplify $\frac{12x^2}{4x}$

(Total for question = 1 mark)

Practice question written for this worksheet (not from a past paper)

2 Simplify $\frac{x^2 + 3x}{x}$

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

3 Simplify $\frac{5x + 10}{x + 2}$

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

4 Write $\frac{x+1}{2} + \frac{x+2}{3}$ as a single fraction.

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

A+ Maths Hub International Community School

5 Write $\frac{2x-1}{3} - \frac{x+4}{6}$ as a single fraction in its simplest form.

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

6 Write $\frac{1}{x} + \frac{1}{y}$ as a single fraction.

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

7 Simplify $\frac{6x^2y}{9xy^2}$

(Total for question = 3 marks)

Practice question written for this worksheet (not from a past paper)

8 Simplify $\frac{x^2-y^2}{3x+3y}$

(Total for question = 3 marks)

Practice question written for this worksheet (not from a past paper)

A+ Maths Hub International Community Schools

9 Express $\frac{3}{4}+\frac{5-x}{6x}$ as a single fraction in its simplest terms.

.....

(Total for question = 3 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2023 · Question 11

10 Express $\frac{5}{4}+\frac{x-3}{6x}$ as a single fraction in its simplest form.

.....

(Total for question = 3 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2025 · Question 12

11 Simplify $\frac{x^2-25}{2x^2-9x-5}$

.....

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · June 2016 · Question 14

A+ Maths Hub International Community Schools

12 Simplify fully $\dfrac{4x^2-25}{6x^2+13x-5}$

.....

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · January 2013 · Question 17

13 Simplify fully $\dfrac{4x^2-9}{2x^2-5x-12}$

Show clear algebraic working.

.....

(Total for question = 3 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2026 · Question 18

14 Simplify fully $\dfrac{4}{x}+\dfrac{3}{2-x}$

.....

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · January 2012 · Question 20

A+ Maths Hub International Community Schools

15 Simplify fully $\dfrac{6x^3 + 13x^2 - 5x}{4x^2 - 25}$

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2019 · Question 22

16 Express $\left(\dfrac{20}{x^2 - 36} - \dfrac{2}{x - 6}\right) \times \dfrac{1}{4 - x}$ as a single fraction in its simplest form.

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2022 · Question 23

17 Write $\dfrac{25x^2 - 64}{5x^2 - 13x - 6} \times \dfrac{x^2 - 8x + 15}{5x + 8} - (x - 7)$

as a single fraction in its simplest form.
Show clear algebraic working.

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · January 2021 · Question 21

A* Maths Hub
International Community Schools

18 Express $\dfrac{4x^2 - 25}{5x^2 + 2x - 7} \times \left(\dfrac{2}{x - 3} - \dfrac{3}{2x - 5} \right)$ as a single fraction in its simplest form.

.....

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2018 · Question 22

19 **23** $\dfrac{4x^2 - 4x - 120}{5x^2 - 180} \div \dfrac{x^2 + 5x}{10x^2 + 60x} = p$ where p is an integer.

Find the value of p

Show clear algebraic working.

$p =$ .....

(Total for Question 23 is 4 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2025 · Question 23

A+ Maths Hub International Community Schools

20 Show that $\dfrac{16x^2 - 36}{x - 7} \div \dfrac{2x^2 + 7x + 6}{x^2 - 5x - 14} - (7 + 8x) = n$

where n is an integer to be found.
Show clear algebraic working.

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2024 · Question 23

21 Write $5 - (x + 2) \div \left(\dfrac{x^2 - 4}{x - 3}\right)$ as a single fraction.

Simplify your answer fully.

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · January 2015 · Question 23

A+ Maths Hub
International Community Schools

22 Express

$$\left(\frac{4}{2x-5}-\frac{3}{2x-3}\right)\div\frac{9x-4x^3}{6x^2-17x+5}$$

as a single fraction in its simplest form.

.....

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · January 2020 · Question 24

23 Write $4-\left[(3x-5)\div\frac{3x^2+x-10}{4x-1}\right]$ as a single fraction in its simplest form.

.....

(Total for Question 26 is 4 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2024 · Question 26

A+ Maths Hub
International Community Schools

24 Write

$$\frac{4x^2 - 17x - 15}{2x - 1} \times \frac{2x^2 - 7x + 3}{x^2 - 25} + (29 - 4x)$$

as a single fraction in its simplest form.

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · January 2023 · Question 26

25 (a) Simplify fully $\left(\frac{256x^{20}}{y^8}\right)^{\frac{1}{4}}$

.....

(2)

(b) Express $\frac{1}{9x^2 - 25} - \frac{1}{6x + 10}$ as a single fraction in its simplest form.

.....

(3)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2018 · Question 15

A+ Maths Hub International Community Schools

26 Express

$$\frac{1}{3x-2} \times \frac{9x^2-4}{3x^2-13x-10} - \frac{7}{x-1}$$

as a single fraction in its simplest form.

.....

(Total for question = 5 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2020 · Question 21

CHALLENGE QUESTIONS

27 Simplify fully $\frac{5}{2x+12} - \frac{x-15}{x^2-2x-48}$

Show clear algebraic working.

.....

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · June 2017 · Question 22

A+ Maths Hub International Community Schools

28 Simplify $(x^2 - 4) \div \left(\frac{4x^2 - 7x - 2}{x}\right) - 2x$

Give your answer in the form $\frac{ax^2}{bx + c}$ where a, b and c are integers.

.....

(Total for question = 4 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2023 · Question 23

29 (a) Factorise $9x^2 - 4y^2$

.....

(2)

(b) Express $\frac{7}{8} - \frac{x + 3}{4x}$ as a single fraction in its simplest form.

.....

(3)

(Total for question = 5 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2022 · Question 11

A+ Maths Hub International Community Schools

30 (a) Express $\frac{5}{8} - \frac{x+2}{4x}$ as a single fraction in its simplest form.

.....

(3)

(b) Expand and simplify $3x(2x - 1)(x + 4)$

.....

(3)

(Total for question = 6 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2026 · Question 16

TOTAL FOR PAPER IS 103 MARKS

A+ Maths Hub International Community Schools